PROGRAM LOGIC AND FLOWCHART

1. Program Logic

Meaning

Problem ko Solve karne ki Soch.

---

2. Algorithm

Definition

Problem Solve karne ka

Step by Step Procedure.

Example

Start

↓

Input

↓

Process

↓

Output

↓

Stop

---

3. Flowchart

Definition

Algorithm ka Graphical Representation.

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4. Flowchart Symbols

Symbol	Work
Oval	Start / Stop
Rectangle	Process
Parallelogram	Input / Output
Diamond	Decision
Arrow	Flow Direction
Circle	On Page Connector
Pentagon	Off Page Connector
Hexagon	Loop

---

5. Programming Pattern

Ye sabse Important hai.

Problem

↓

Algorithm

↓

Flowchart

↓

Python Code

↓

Output

---

6. Common Flow

Har Basic Program

START

↓

INPUT

↓

PROCESS

↓

OUTPUT

↓

STOP

Isi Pattern ko Follow karta hai.

---

7. Programs Jo Humne Sikhe

☑ Sum of Two Numbers

☑ Area of Circle

☑ Simple Interest

Sabka Structure Same tha.

🧠 Python Insight

Ab tum notice karo.

Python me hum jo Program likhenge,

Unka bhi Pattern yahi rahega.

Example

```
# Input
a = int(input())
b = int(input())

# Process
sum = a + b

# Output
print(sum)
```

Ye Flowchart ka hi Code Version hai.

🤖 AI/ML Connection

Machine Learning Model bhi isi Process ko Follow karta hai.

Problem

↓

Collect Data

↓

Train Model

↓

Prediction

↓

Result

Har AI System ke piche bhi ek Algorithm hota hai.

Isliye Chapter 2 AI/ML ke liye bhi bahut important tha.

🎯 Chapter 2 se Kya Yaad Rakhna Hai?

Formula yaad nahi...

Pattern yaad rakho.

INPUT

↓

PROCESS

↓

OUTPUT

Aur

Algorithm

↓

Flowchart

↓

Python Program

Bas.

Ye hi Programming ka Base hai.

1. Program Logic

Program Logic kisi problem ko solve karne ke liye sahi soch aur sahi sequence of steps ko kehte hain.

Example

Tea banana.

Water lo

↓

Gas ON karo

↓

Tea Patti dalo

↓

Milk dalo

↓

Tea Ready

Ye hi Program Logic hai.

---

2. Problem Solving Methodology

Programming me problem solve karne ke steps.

Problem

↓

Analysis

↓

Algorithm

↓

Flowchart

↓

Coding

↓

Testing

↓

Final Program

---

3. Algorithm

Algorithm ek Step-by-Step Procedure hai jo kisi problem ko solve karta hai.

Characteristics

- Clear Steps
- Proper Sequence
- Easy to Understand
- Finite (End hona chahiye)

Example

Start

↓

Input

↓

Process

↓

Output

↓

Stop

---

4. Flowchart

Flowchart Algorithm ka Graphical Representation hota hai.

Ye Symbols ka use karke Program ki working dikhata hai.

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5. Advantages of Flowchart

- Program samajhna easy hota hai.
- Planning improve hoti hai.
- Errors kam hote hain.
- Team work easy hota hai.
- Documentation me help karta hai.
- Program modify karna easy hota hai.

---

6. Flowchart Symbols

Terminator

Start aur Stop ko dikhata hai.

Shape

Oval

Process

Calculation ya Action.

Shape

Rectangle

Input / Output

Data lena ya Result dikhana.

Shape

Parallelogram

Decision

Yes / No Decision.

Shape

Diamond

Line Connector

Flow Direction dikhata hai.

Shape

Arrow

On Page Connector

Same page par Flow continue karta hai.

Shape

Circle

Off Page Connector

Next page par Flow continue karta hai.

Shape

Pentagon

Loop

Same work ko Repeat karta hai.

Shape

Hexagon

---

7. Programming Pattern

Har Basic Program isi Pattern ko Follow karta hai.

Start

↓

Input

↓

Process

↓

Output

↓

Stop

---

8. Algorithm Examples

Sum of Two Numbers

Start

↓

Input A, B

↓

Sum = A + B

↓

Print Sum

↓

Stop

Area of Circle

Formula

Area = πr^2

Algorithm

Start

↓

Input Radius

↓

Area = π × r^2

↓

Print Area

↓

Stop

Simple Interest

Formula

SI = (P × R × N)/100

Algorithm

Start

↓

Input P,R,N

↓

SI = (P × R × N)/100

↓

Print SI

↓

Stop

---

9. Python Programming Pattern

Input

↓

Calculation

↓

Output

Example

```
a = int(input())
b = int(input())
sum = a + b
print(sum)
```

---

10. AI/ML Connection

AI Projects bhi isi Logic ko Follow karte hain.

Input Data

↓

Processing

↓

Prediction

↓

Output

⭐ Important Formula

Area of Circle

Area = πr^2

Simple Interest

SI = (P × R × N)/100

📖 Important Definitions (Exam Ready)

1. Program Logic

Program Logic is the sequence of logical steps used to solve a problem.

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2. Algorithm

An Algorithm is a step-by-step procedure used to solve a problem.

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3. Flowchart

A Flowchart is a graphical representation of an algorithm using standard symbols.

---

4. Terminator

A Terminator symbol is used to indicate the Start and End of a program.

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5. Process

A Process symbol represents calculations or processing steps in a program.

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6. Input / Output

The Input/Output symbol is used to receive data from the user or display the result.

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7. Decision

A Decision symbol is used to check a condition and choose between Yes or No paths.

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8. Line Connector

A Line Connector shows the direction of program execution in a flowchart.

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9. On-Page Connector

An On-Page Connector connects different parts of a flowchart on the same page.

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10. Off-Page Connector

An Off-Page Connector connects a flowchart to another page.

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11. Loop (Repetition)

A Loop is used to repeat a set of instructions until a condition is satisfied.

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12. Problem Solving Methodology

Problem Solving Methodology is a systematic approach used to analyze, design, develop, test and implement a solution.

🧠 Memory Trick

Programming ka Golden Rule:

Problem

↓

Algorithm

↓

Flowchart

↓

Python Code

↓

Output

Aur har Program ka Basic Structure:

START

↓

INPUT

↓

PROCESS

↓

OUTPUT

↓

STOP